

src/components/CodeCityView.tsx

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Overview

`src/components/CodeCityView.tsx` (added) implements a 3-D view of code cities with Three.js. It replaces the previous placeholder and exposes a React component that renders the city, handles user interaction, and cleans up on unmount.

Key changes

- **Imports** (lines 1-3): `useEffect`, `useRef` from React; `three` core; `OrbitControls` from `three/examples/jsm/controls/OrbitControls.js`.
- **Type imports** (lines 5-6): `CodeCityLayoutResult`, `CodeCityRenderableBuilding` from `../app/codeCityLayout`.
- **`attachSelectable` helper** (lines 9-14): recursively assigns `object.userData.building` to every child of a building group, enabling hit-testing.
- **`CodeCityView` component** (lines 16-263):
 - Builds a Three.js scene, camera, renderer, ambient and directional lights, a grid helper, and meshes for districts and buildings.
 - Uses `OrbitControls` for navigation and a `Raycaster` to detect clicks on building meshes, calling the supplied `onSelect`.
 - Observes the container with `ResizeObserver` to keep camera aspect and renderer size in sync.
 - Disposes all Three.js objects and removes event listeners on unmount.
- **UI fallback** (lines 218-225): shows a friendly message when `layout.buildings` is empty.

Impact

- Provides a fully interactive 3-D rendering of the city layout.
- Centralizes Three.js setup in a single component, keeping mesh-tagging logic isolated via `attachSelectable`.
- Uses `requestAnimationFrame` for continuous rendering and `OrbitControls` damping for smooth navigation.
- No new global state or external APIs are introduced.

Risks & follow-ups

- **Memory leaks:** `dispose()` is called on renderer, controls, and scene, but `attachSelectable` may leave references; verify cleanup.
- **Resize handling:** `ResizeObserver` is used; test across browsers, especially on very small or large containers.
- **Raycaster precision:** nested meshes rely on `userData.building`; confirm that deeply nested buildings are correctly detected.

- **Compatibility:** importing from `three/examples/jsm/...` may fail in older bundlers; run the production build to confirm.